



Employees' reskilling and upskilling for industry 5.0: Selecting the best professional development programmes

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ABSTRACT

The current research aims at developing a multi-criteria decision-making methodology, based on analytic network process (ANP), for selecting the appropriate professional development programmes for enhancing employees' reskilling and upskilling, in the Industry 5.0 context. Thus, a four-stage methodology, which combines the qualitative approach of the Delphi study with the quantitative approach of the multi-criteria decision-making technique of ANP, is proposed and tested in the three largest companies from the Romanian manufacture of coke and refined petroleum products industry. The results show that teamwork, online communities of practice and enterprise social networks are among the best options. Furthermore, the most influential factors are employees' psychological profile (namely, interpersonal trust and self-efficacy) and human capital (especially, inclusiveness and job satisfaction). These findings have both theoretical and practical implications. At the theoretical level, they provide a nexus between human resource management and operations management by showing how a methodology based on ANP can be applied for solving a human resource issue. At the practical level, they help managers to decide which professional development programmes to implement for employees' reskilling and upskilling by providing them an easily customizable tool that takes into account employees' characteristics and company's human and structural capital.

1. Introduction

The concept of Industry 5.0 has recently been adopted by the [1] in order to highlight the need for developing human-centric, sustainable and resilient industries. Unlike the previous industrial revolutions which aimed to increase productivity by eliminating or alienating humans [2], Industry 5.0 places humans back at the center of attention and presents them as the most important organizational resources [3]. Thus, while Industry 4.0 puts a focus on digital transformation and highlights how process efficiency can be increased by using various digital technologies, Industry 5.0 focuses on the creative thinking skills of humans and presents the human-machine collaboration as the key to ensure industries flexibility and agility [2]. The focus is no longer on process automation like in the Industry 4.0 context but on creating agile, adaptable, and sustainable industries by using technologies and tools to empower and promote employees' talents, skills and competences (De Nul, 2021; [4, 5]). However, it must be stated that Industry 5.0 connects to Industry 4.0 applications and technologies, but these are treated as a mean to promote mass personalization and not as a purpose.

Although it is a relatively new concept and various scholars treat it as

a future event (Modgli et al., 2023 [2–4]; state that Industry 5.0 is already in practice in both service and production sectors, and one of the most prominent barriers in its development is represented by employees reskilling and upskilling. Hence, roughly 30–40 % of employees will have to upgrade significantly their skills, in the next ten years [6] since two-thirds of skills that are critical for today's job requirements will change or become obsolete generating a skills gap with a potential economic effect of 2.5tn \$ [7,8]; I4MS, 2022; [9]. Thus, jobs like bank clerks, cashiers, data entry clerks, administrative and executive secretaries, accounting, bookkeeping, and payroll clerks, security guards, software testers, that require the use of physical and routine skills (such as, manual dexterity, endurance and precision, data entry, data management, and project management) are expected to disappear since tasks that involve their use will be automatically performed by robots and machines [10,11]. On the other hand, new jobs are created, like sustainability specialists, talent acquisition associate, customer success analyst, growth marketing manager, workplace coordinator, that involve the use of cognitive skills [11]. According to the Future Jobs Report 2023, the core skills required for the development of Industry 5.0 include: creative thinking (73.20 %), analytical thinking (71.60 %),

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curiosity and lifelong learning (66.80 %), resilience, flexibility, and agility (65.80 %) and systems thinking (59.90 %). These emphasize that, against the Industry 5.0 backdrop, employees will have to make accurate and less unstable decisions, not just performing certain tasks, while dealing with an increased speed at which they are getting information. Therefore, they will have to be able to understand complex situations, to find innovative solutions and to adapt to a continuously changing environment.

Even though there is a growing concern for identifying which skills should be targeted by companies' reskilling and upskilling strategies [10,12–14], there is no published study indicating which professional development programs could be used. Their importance is highlighted by Refs. [15,16] from a general perspective but, as [17] state, it is unclear what kind of professional development programs should be implemented in order to respond to the forthcoming challenges generated by Industry 5.0. This study will fill this gap by proposing and testing a methodology for selecting the appropriate professional development programs. As a consequence, it will answer to employers' concern for reskilling and upskilling their employees by providing a viable tool for decision-making.

Furthermore, among the studies that analyze the professional development programs [18–21], two lines of research can be identified. The first one concentrates on emphasizing how professional development programs, such as mentoring, training, enterprise social networks, serious games, volunteering activities etc., foster skills development while the second one highlights the individual and organizational factors that may influence their efficiency. The perspective adopted by the scholars from the second category is limited by the fact that they neglect the interdependency that exists within and between individual and organizational factors. For example [22,23], claim that mentoring efficiency depends on commitment and organizational culture, but they do not consider the fact that organizational culture moderates the relationship between commitment and interpersonal trust [24]. By taken into account the dependency established among the influence factors, a viable decision-making tool could be developed.

Based on the aforementioned gaps, the current study aims to answer to the following questions research question:

RQ: Which is the most appropriate professional development program for employees' reskilling and upskilling, in the Industry 5.0 context, considering the individual and organizational influence factors?

Hence, the goal of this research is to propose and test a multi-criteria methodology for prioritizing and selecting the most appropriate professional development program that supports employees' reskilling and upskilling, in the Industry 5.0 context.

The reminder of the paper is structured as follows. Section 2 reviews the relevant scientific literature about the skills required in the Industry 5.0 and the professional development programs that could facilitate their development. It also emphasizes the main research gaps and the characteristics of the proposed model. Section 3 presents the research goal and the methodology developed for achieving this goal while Section 4 synthesizes the main results. Section 5 highlights the main theoretical and practical implications of the research findings while Section 6 closes this article by providing directions for future research avenues regarding reskilling and upskilling in the context of Industry 5.0.

2. Theoretical framework

2.1. Requisite skill set for industry 5.0

The characteristics and content of jobs changed significantly in the past years, and so did the skill set that the employers are looking for. Thus, the 3 R s (Reading, wRiting, aRithmetic) were replaced by the 3 S s (Searching, Sharing, Simulation) in the light of the technological changes. Taking into account the polarization approach, according to

which a decline in employment in mid-skilled occupations and a growth in low- and high-skilled ones is expected [25–27], Grdr Broo et al. (2022) state that the future labor market is defined by the 4Cs (Critical thinking, Creativity, Collaboration, Communication). This set of skills is highly valued in the Industry 5.0 since it facilitates mass personalization. Based on collaboration and communication, employees get closer to the customers, understand better their expectations and develop better versions of customer experience maps [28,29], and enhance the implementation of co-creation and co-design activities [4,30]. Based on critical thinking and creativity, they are able to solve complex problems, define innovative solutions, explore new business paths and create extra-value [4]; Scuotto et al., 2023). Furthermore, this set of skills sheds light on the unique value of individuals, the only ones capable of transforming all the other organizational resources and generating value-added, since the tasks that nowadays require the use of routine skills, like data entry, basic statistics, database management, and project management, will be automatically performed by robots and machines [10]. In other words, in the Industry 5.0, employees use and complement digital technologies; they need critical thinking and creativity to go beyond the information provided by the digital technologies, and collaboration and communication in order to know how and when to use digital technologies.

Nevertheless, while some scholars concentrate on the requisite skill set for Industry 5.0 [31], go further and argue that there will be four types of jobs on the market, namely.

- technical jobs with low complexity that involve routine tasks and demand a low-complexity skill set and an automated decision-making process;
- social jobs with low complexity that involve routine tasks and human-to-human interaction, and demand social skills like trust, compassion, and empathy;
- complex technical jobs that involve human-to-machine interaction and unstructured decision-making processes, and demand technical skills and creativity;
- complex social jobs that involve social interactivity and demand soft skills like social skills (compassion, trust, empathy), human skills (communication, self-awareness, listening), creativity and problem-solving.

Hence, employees' reskilling and upskilling is required at the soft skills level in order to ensure company's competitiveness in the Industry 5.0 context.

2.2. Effectiveness of professional development programs

According to Ref. [15]; reskilling means employees need the knowledge and skills to take on different roles while upskilling means the employees gain new skills which can improve their performance on their current jobs. The professional development programs answer specifically to this requirement since their objective is to improve employees' knowledge and skills in order to increase their achievements. Still, not all the professional development programs facilitate the development of the same set of skills.

The specialized literature analyzing the classical professional development programs point out that apprenticeship is one of the major means to transfer skills [21] while mentoring turns out to be highly efficient for developing employees' problem-solving ability and innovation skills [19]. Teamwork enhances creativity [32] and the development of social skills [33] while training fosters the development of digital skills [20], and also cognitive and interpersonal skills [34]. Furthermore, traditional and digital storytelling have an impact not only on the development of information gathering skills and problem-solving [35] but also on creativity and critical thinking [36]. The modern professional development programs, such as enterprise social networks and online communities of practice, have been successfully used for

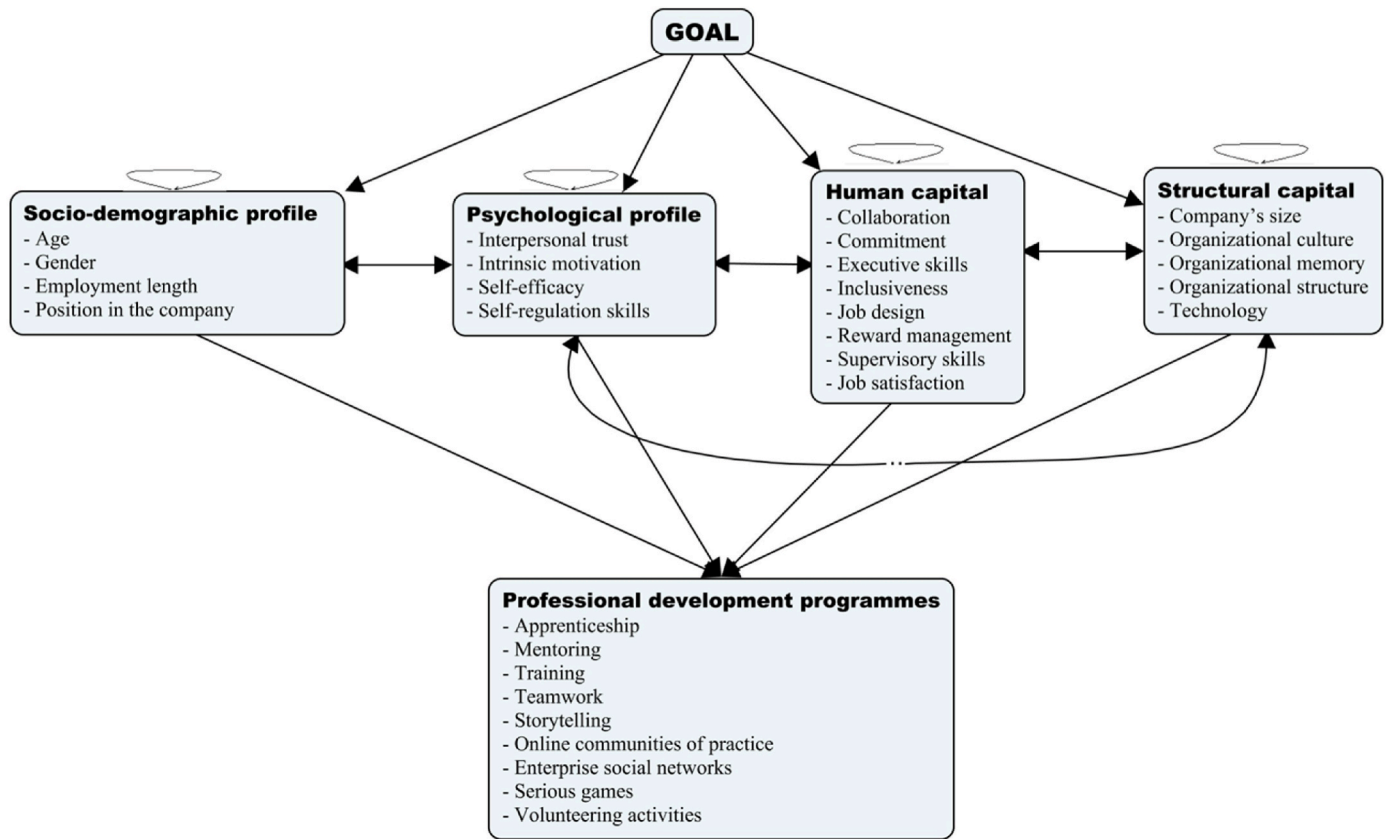


Fig. 1. The components of the ANP.

developing employees' critical thinking [37], digital skills [38], and social skills [39]. Moreover, serious games are used for developing employees' soft skills [40,41] while volunteering activities, which represent one of the fast-growing trends in skills development [42], enhance innovation, creativity, empathy, and problem-solving.

Additionally, it must be mentioned that 85 % of the Fortune 500 companies use at least one type of professional development programs for employees' reskilling and upskilling [43]. Thus, mentoring is used by Cartepillar and General Electric for facilitating continuous learning while Boeing uses it for enhancing the development of communication skills. Teamwork is used by Google for developing collaboration while Tesla and Amazon use it for fostering innovation and intrapreneurship. Last but not least, Marriott implements storytelling for developing employees' creativity while Axero Solutions pursues the same objective by using enterprise social networks.

Still, no information is provided on which is the best choice for preparing employees for Industry 5.0 challenges. This information is crucial since managers are always confronted with limited resources and unlimited needs, and they are reluctant to experiments (trial and error situations) when they know that a mistake can cost their company time and money [8].

2.3. Proposed model

As previously mentioned, the studies developed so far emphasize how the professional development programs can enhance the development of soft skills, but they don't provide a viable tool that could help managers decide which program to implement. This involves a multi-criteria decision since managers have to consider both individual factors, related to employees' socio-demographic and psychological profile, and organizational factors, which usually describe company's human and structural capital.

The specialized literature regarding professional development programs adopts two perspectives; some scholars focus on individual factors while others focus on the organizational factors. Those from the first line of research show that the effectiveness of the professional development programs depends on employees' socio-demographical and psychological factors. Among the socio-demographical factors, the researchers (Brumeister et al., 2018 [44]; Geeraerts et al., 2018 [45]; point out the influence of age, gender, employment length, and position in the company. Regarding the psychological factors, previous studies [46,47]; Nisula & Metso, 2019) highlight the influence of interpersonal trust, intrinsic motivation, self-efficacy, and self-regulation skills.

The scholars from the second line of research concentrate on the organizational factors and shed light on the fact that the effectiveness of the professional development programs depends on two components of intellectual capital, namely human capital and structural capital. The factors included in the human capital category are collaboration, commitment, executive skills, inclusiveness, job design, reward management, supervisory skills, and job satisfaction [48,49]; Sirca et al., 2013). Besides, the scholars that focus on the structural capital category [16,22,23,50] bring forward the influence of company's size, organizational culture, organizational structure, organizational memory, and technology.

Despite the fact that these studies bring forward the factors that should be considered by the policy-makers when deciding which program to implement, they omit the fact that these factors are not independent. Several studies from the human resource management area show that various interdependencies exist within and between individual and organizational factors. Thus [51], show that interpersonal trust is critical in determining all aspects of collaboration while [52] state that it has a significant impact on commitment. Furthermore [24], argue that organizational culture moderates the relationship between interpersonal trust and commitment while [53] reveal that commitment



Fig. 2. The proposed four-stage methodology.

mediates the relationship between organizational culture and employee intention to stay [54], highlight the moderate effect of age on the relationship between rewards and job satisfaction while [55] argues that reward management and trust influence the organizational culture. Given the high complexity of the decision that the managers have to make when deciding which program to implement for employees' reskilling and upskilling, a multi-criteria decision-making model is required. The current research fills this gap by proposing and testing a multi-criteria methodology, based on ANP, for prioritizing and selecting the professional development programs that support employees' reskilling and upskilling.

ANP is the most popular multi-criteria decision-making technique [56] due to the fact that it converts the qualitative components into quantitative elements for the relative analysis of preferences and it allows correlations within and between groups of components [57]. In other words, it is capable of representing a real-world decision problem using loose network structure [58], and to bring forward the decision-ranking priorities. Due to its ability to convert the interdependence established among the decisional elements into priority vectors, it was successfully applied in the operations management area for suppliers' selection [59,60], performance evaluation [61], and sustainability evaluation (Lin et al., 2021).

Taking this into account and the results provided by the specialized literature, the structure of the analytic network is defined (Fig. 1). As it can be noticed, at the first level, the overall objective is established namely, selecting the appropriate professional development programs for fostering employees' reskilling and upskilling. At the second level, the influence factors and the relationships established among them are considered. Thus, the influence factors are grouped in four clusters, as defined in the specialized literature, namely: employees' socio-demographical profile, employees' psychological profile, company's human capital, and company's structural capital. Each cluster includes the factors that previous studies have highlighted as having an influence on the effectiveness of the professional development programs. Further, based on the human resource management literature, the relationships established among factors are highlighted. The interdependencies are marked by arrows and the inner-dependencies by looped arc; in other words, the relationships established between the factors that belong to different clusters are reflected by arrows while the relationships established between the factors that belong to the same cluster are emphasized by looped arc. At the last level of the proposed model, the alternatives are presented; these include the professional development programs that can be used for employees' reskilling and upskilling, according to the specialized literature. The characteristics of the testing phase and the generated results are highlighted further.

3. Research design

3.1. Methodological approach

This study aims to develop a multi-criteria methodology, based on ANP, for selecting the appropriate professional development programs for fostering employees' reskilling and upskilling in the Industry 5.0 context. A four-stage methodology is developed (Fig. 2); the output of one stage serves as an input for the following stage.

Stage 1: Questionnaire design. Starting from the model presented in Fig. 1, a questionnaire is developed. This includes 32 questions, organized in six sections. Section one includes 6 questions focusing on evaluating the importance of the four categories of factors (namely,

Table 1

The nine-point priority scale of [58].

Value	Level of importance	Description
1	Equal	Two variables are equally important.
2	Weak or slight	One variable is equally to moderately more important than the other.
3	Moderate	One variable is moderately more important than the other.
4	Moderate plus	One variable is moderately to strongly more important than the other.
5	Strong	One variable is strongly more important than the other.
6	Strong plus	One variable is strongly to very strongly more important than the other.
7	Very strong	One variable is very strongly more important than the other.
8	Very, very strong	One variable is very strongly to extremely more important than the other.
9	Extreme	One variable is extremely more important than the other.

socio-demographic profile, psychological profile, human capital, and structural capital) in selecting the professional development programs. Section two concentrates on the socio-demographic profile and includes 4 questions. Section three focuses on the psychological profile and includes 4 questions. Section four reunites 8 questions concerning human capital. Section five includes 5 questions about structural capital while section six concentrates on respondents' socio-demographical characteristics, which are captured by 5 questions. It must be mentioned that the questions included in section one to five are developed using a pair-wise comparison matrix that involves comparing the professional development programs with regard to a specific criterion. For example, the respondents have to answer to the following question: "With regards to the employees' interpersonal trust, how important is apprenticeship compared with mentoring?". For this type of questions [58], scale of 1–9 (Table 1) is used due to the fact that it "indicates how many times more important or dominant one element is over another element with respect to the criterion or property with respect to which they are compared" (Saaty, 2008, p.85) and it has successfully been applied in previous studies that involved pair-wise comparison [62,63].

Stage 2: Delphi study. Survey based on a questionnaire. A Delphi study is employed in order to capture the collective opinion and expertise of the main players on the Romanian market of manufacture of coke and refined petroleum products while ensuring respondents quasi-anonymity. According to Rauch (1979, p.164), this is able "to coordinate and structure the general lines of thinking in a different and unexplored field of social relations and to transfer the future development of such an area from mere accident to carefully considered decisions". Furthermore, Delphi study is frequently used in the human resource research [64]; Ruiz-Perez et al., 2022 [65]; since it is able to provide valuable insights on intangible issues.

Thus, in the first round, the questionnaires are distributed among the national and regional human resource managers from three of the largest companies from the Romanian industry of manufacture of coke and refined petroleum products. Data are collected from 13 human resource managers who have more than (i) 15 years of experience in human resource management, and (ii) 5 years of experience in the company. The results are analyzed, and several differences are identified regarding the importance of gender, reward management, and

organizational memory.

As a consequence, a second round of questionnaires is sent together with the consolidated results of the first round. The respondents are invited to review their answers and to adjust their evaluation if they want to. At the end of the second round, data are collected and analyzed, and it is noticed that consensus has been reached and consistency is

$$S = \begin{matrix} & \begin{matrix} C_1 & C_2 & \dots & C_N \end{matrix} \\ \begin{matrix} C_1 \\ C_2 \\ \vdots \\ C_N \end{matrix} & \begin{bmatrix} \begin{matrix} e_{11} & e_{12} \dots e_{1n_1} \\ e_{21} & e_{22} \dots e_{2n_2} \\ \vdots & \vdots \\ e_{N1} & e_{N2} \dots e_{Nn_N} \end{matrix} & \begin{matrix} W_{11} & W_{12} & \dots & W_{1N} \\ W_{21} & W_{22} & \dots & W_{2N} \\ \vdots & \vdots & \dots & \vdots \\ W_{N1} & W_{N2} & \dots & W_{NN} \end{matrix} \end{bmatrix} \end{matrix} \quad (5)$$

ensured.

Stage 3: Data processing using ANP. The data collected during the second round of the Delphi study are processed using an ANP approach. ANP is a multi-criteria decision-making technique that takes into account the interdependence established among the decisional elements (Teng et al., 2012) and converts it into priority vectors. It was successfully used for suppliers' selection [59,60], performance evaluation [61], and sustainability evaluation (Lin et al., 2021), due to the fact that, based on factors' importance and the relationships established among them, it ranks the alternatives. Hence, in order to determine the most appropriate professional development programs for employees' reskilling and upskilling, the following steps are followed:

Step 1. Based on the data collected through Delphi study, using Saaty's 1 to 9 scale, a comparison matrix is created for each factor and alternative. For example, for the factor interpersonal trust, the matrix reflects how appropriate each professional development program is compared with all the other ones. The pairwise comparison matrices are constructed by applying the following equation:

$$A = [a_{ij}] = \begin{bmatrix} 1 & a_{12} & \dots & a_{1n} \\ a_{21} & 1 & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & 1 \end{bmatrix} \quad (1)$$

where a_{ij} reflects the relative importance of the variable in the i th row compared to that in the j th column.

Step 2. The eigenvectors are determined, based on the following equation:

$$W = \left(\prod_{j=1}^n a_{ij} \right)^{1/n} / \sum_{i=1}^n \left(\prod_{j=1}^n a_{ij} \right)^{1/n} \quad (2)$$

where n is the number of variables being compared in the matrix.

Step 3. The consistency ratio (CR) is estimated, based on:

$$CR = \frac{CI}{RI} \quad (3)$$

where: CI is the consistency index of the pairwise comparison;

RI is the random consistency index.

The consistency index of the pairwise comparison (CI) is obtained by:

$$CI = \frac{\lambda_{\max} - n}{n - 1} \quad (4)$$

The consistency ratio of the developed pairwise comparison matrices ranges from 0 to 0.09325. Given the fact that its value is less than 0.1 [66], data are considered to be consistent.

Step 4. The unweighted supermatrix is developed, as follows: where: C_n is the n th cluster;

E_{nm} is the m th element in the n th cluster;

W_{ij} is the priority vector of the influence elements compared in the j th cluster to the i th cluster.

Step 5. The unweighted supermatrix is transformed into a weighted supermatrix by factorizing the elements of each column based on their relative weight. Hence, all the columns are sum to unity, and the weighted supermatrix becomes a stochastic matrix.

Step 6. The weighted supermatrix is raised to limiting powers in order to obtain the global priorities and then, the alternatives are ranked from best (the one that has the highest importance coefficient) to worst (the one that has the lowest importance coefficient). The following equation is applied:

$$s = \lim_{k \rightarrow \infty} S^k \quad (6)$$

In other words, the values obtained previously are normalized in order to sum 1 by dividing the relative priority of each professional development program by the sum of all the relative priority vectors. Further, to highlight the most appropriate professional development program, each normalized value is divided by the largest one.

Stage 4: Data analysis. Data are analyzed and sensitivity analysis is performed in order to evaluate the evolving scenario [67]. This emphasizes which factors could generate a significant change in the ranking with the least variation [68].

3.2. Participants

The analysis focuses on the Romanian industry of manufacture of coke and refined petroleum products due to the fact that, in 2021, Romania was the 39th largest exporter of petroleum coke in the world, and based on market concentration, its main exporting competitors are represented by United States, Canada and Singapore, countries that concentrate on increasing processes digitalization [69]. Furthermore, the industry is dominated by large companies with high levels of value-added and, as [70] state, it moves faster to the Industry 5.0.

Against this backdrop, the three most important players on the market are selected based on their reputation, their employment capacity, and also the fact that they act as a factor of change at the industry level, setting the tendencies (they serve as a model for the other companies on the market). The human resource managers of these companies are contacted and invited to be part of the research. They accepted the invitation and provided the contact details of a total of 10 persons who work within the company as regional human resource managers. Thus, using a snowball sampling technique, a sample of 13

Table 2
Respondents' socio-demographic characteristics.

Company Characteristics	A	B	C
<i>Gender</i>			
Masculine	3	1	
Feminine	2	4	3
<i>Age</i>			
<25 years			
26–35 years			
36–45 years	2	1	2
>45 years	3	4	1
<i>Experience in human resource management</i>			
<1 year			
1–3 years			
3–5 years			
5–10 years			
10–15 years			
>15 years	5	5	3
<i>Experience in implementing professional development programs</i>			
<1 year			
1–3 years		1	
3–5 years			1
5–10 years	2	1	
10–15 years		1	2
>15 years	3	2	
<i>Experience in the company</i>			
<1 year			
1–3 years			
3–5 years			
5–10 years	1	3	2
>10 years	4	2	1

national and regional human resource managers is obtained. Their socio-demographical characteristics are presented in Table 2. As it can be noticed, they have more than 15 years of experience in the human resource management field and work for the company for more than 5 years.

4. Results

4.1. Prioritization of the professional development programs

The data obtained through ANP approach bring forward the most important factors that are taken into consideration when selecting the professional development programs that are going to be implemented for employees' reskilling and upskilling. As it can be observed from

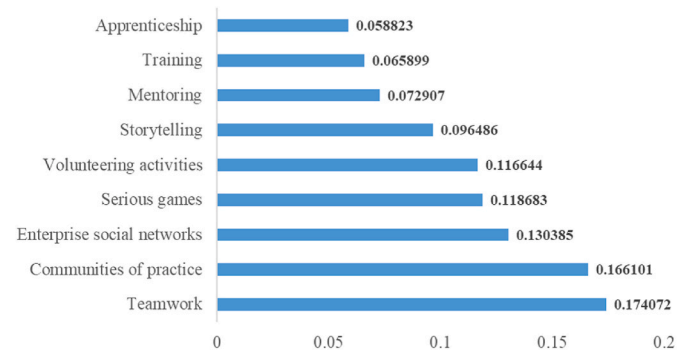


Fig. 3. The priority vector of the professional development programs.

Table 3, the focus is on individual factors, which have a priority vector three times higher than organizational factors. Employees' psychological profile (with a priority vector of 0.656) is crucial for employees' reskilling and upskilling, especially due to their interpersonal trust (with a priority vector of 0.389) and self-efficacy (with a priority vector of 0.389). Both elements tend to influence employees' openness and interest in developing their skills, and they also reflect the fact that managers perceive reskilling and upskilling as hyper-personalized activities. On the other hand, among the organizational factors, inclusiveness (with a priority vector of 0.277) and job satisfaction (with a priority vector of 0.251) are considered critical. Thus, reskilling and upskilling become a function of employees' needs, and the professional development programs turn to be employee-owned.

As a consequence, the appropriate professional development programs for employees' reskilling and upskilling, in the manufacture of coke and refined petroleum products industry, are represented by teamwork, online communities of practice and enterprise social networks (Fig. 3); these allow for reskilling and upskilling to be hyper-personalized, centered on employees' needs and preferences. While the former supports the development of hard and soft skills by experience, the last two programs not only enhance the development of digital skills, but they also allow employees to freely upskill, based on their needs, desires, and expectations. It can be noticed that the three most appropriate professional development programs are based on socialization and one-to-many communication and support a continuous interchange of explicit and tacit knowledge flows.

Besides, the classical professional development programs based mostly on one-to-one communication and having a certain formal

Table 3
The priority vector of clusters, criteria, and sub-criteria.

Cluster	Priority	Criteria	Priority	Sub-criteria	Priority					
Individual factors	0.75	Socio-demographic profile	0.093	Age	0.164					
				Gender	0.352					
				Employment length	0.342					
		Psychological profile		0.656	Position in the company	0.140				
					Interpersonal trust	0.389				
					Intrinsic motivation	0.152				
					Self-efficiency	0.389				
					Self-regulation skills	0.067				
					Organizational factors	0.25	Human capital	0.208	Collaboration	0.158
									Commitment	0.108
Executive skills	0.038									
Inclusiveness	0.277									
Job design	0.061									
Reward management	0.050									
Supervisory skills	0.053									
Structural capital	0.041	Job satisfaction	0.251							
		Company's size	0.031							
		Organizational culture	0.419							
		Organizational memory	0.298							
		Organizational structure	0.059							
Technology	0.191									

character, such as mentoring, training, and apprenticeship, are considered to be less important.

4.2. Sensitivity analysis

According to the sensitivity analysis, several variations can occur in the ranking if changes appear at the criteria level (Table 4). The most significant variations appear when the changes take place at the level of employees' socio-demographic profile. In this case, the importance of the online communities of practice and enterprise social networks decreases while the importance of mentoring and volunteering activities increases; if a linear increase of 50 % occurs ($\alpha = 0.75$) then teamwork, volunteering activities and communities of practice become the most suitable professional development programs for employees' reskilling and upskilling.

If the changes that occur at the level of employees' psychological profile are considered, it can be noticed that interpersonal trust and self-efficacy inhibit professional development programs while intrinsic motivation and self-regulation skills act as a stimulus (Table 5). Thus, if the importance of interpersonal trust is reduced by 50 % ($\alpha = 0.25$) then the overall change in professional development programs will be equal to 23.05 %. On the other hand, if it increases by 50 % ($\alpha = 0.75$), the overall change will be equal to -30.43 %. Furthermore, enhancing employees' interpersonal trust has a positive effect on those activities based on collaboration and direct interaction, namely: online communities of practice, serious games, and teamwork.

If the weight of importance of intrinsic motivation is reduced by 50 % ($\alpha = 0.25$) then the overall change will be equal to -15.99 %. On the

other hand, if it increases by 50 % ($\alpha = 0.75$), the overall change will be equal to 97.14 %. Moreover, enhancing employees' intrinsic motivation has a positive effect on those activities that support skills development by doing/direct experience, namely: apprenticeship, online communities of practice, mentoring, and training.

Hence, if the weight of importance of self-efficacy is reduced by 50 % ($\alpha = 0.25$) then the overall change will be equal to 9.73 %. On the other hand, if it increases by 50 % ($\alpha = 0.75$), the overall change will be equal to -9.36 %. Therefore, enhancing employees' self-efficacy has a positive effect on those activities involving one-to-many communication and sharing experiences, such as: enterprise social networks, storytelling, and volunteering.

Besides, if the weight of importance of self-regulation skills is reduced by 50 % ($\alpha = 0.25$) then the overall change will be equal to -2.29 %. On the other hand, if it increases by 50 % ($\alpha = 0.75$), the overall change will be equal to 55.51 %. Enhancing employees' self-regulation skills has a positive effect on apprenticeship, serious games, and training.

Similarly, if the analysis focuses on the changes that occur at the level of human capital (Table 6), it can be noticed that professional development programs are enhanced by.

- commitment (the overall change varies from -3.09 %, when $\alpha = 0.25$ -26.30 %, when $\alpha = 0.75$),
- job design (the overall change varies from -4.65 %, when $\alpha = 0.25$ -68.25 %, when $\alpha = 0.75$),
- reward management (the overall change varies from -0.25 %, when $\alpha = 0.25$ -16.62 %, when $\alpha = 0.75$),

Table 4
Sensitivity analysis for the main criteria.

Criteria	Professional Development Programs	Alpha value (α)					
		0.25		0.5		0.75	
		Priority	Rank	Priority	Rank	Priority	Rank
Socio-demographic profile	Apprenticeship	0.056	9	0.059	9	0.085	8
	Communities of practice	0.171	2	0.166	2	0.125	3
	Enterprise social networks	0.134	3	0.130	3	0.103	7
	Mentoring	0.068	7	0.073	7	0.115	4
	Serious games	0.119	4	0.119	4	0.112	5
	Storytelling	0.095	6	0.096	6	0.110	6
	Teamwork	0.178	1	0.174	1	0.141	1
	Training	0.064	8	0.066	8	0.083	9
	Volunteering activities	0.116	5	0.117	5	0.126	2
Psychological profile	Apprenticeship	0.072	9	0.059	9	0.051	9
	Communities of practice	0.141	2	0.166	2	0.181	2
	Enterprise social networks	0.120	3	0.130	3	0.136	3
	Mentoring	0.096	7	0.073	7	0.059	7
	Serious games	0.114	5	0.119	4	0.121	4
	Storytelling	0.098	6	0.096	6	0.096	6
	Teamwork	0.160	1	0.174	1	0.182	1
	Training	0.080	8	0.066	8	0.058	8
	Volunteering activities	0.119	4	0.117	5	0.115	5
Human capital	Apprenticeship	0.057	9	0.059	9	0.066	9
	Communities of practice	0.173	2	0.166	2	0.142	2
	Enterprise social networks	0.132	3	0.130	3	0.126	3
	Mentoring	0.068	7	0.073	7	0.090	7
	Serious games	0.121	4	0.119	4	0.112	5
	Storytelling	0.098	6	0.096	6	0.091	6
	Teamwork	0.175	1	0.174	1	0.172	1
	Training	0.062	8	0.066	8	0.080	8
	Volunteering activities	0.115	5	0.117	5	0.121	4
Structural capital	Apprenticeship	0.058	9	0.059	9	0.068	9
	Communities of practice	0.166	2	0.166	2	0.173	1
	Enterprise social networks	0.130	3	0.130	3	0.135	3
	Mentoring	0.072	7	0.073	7	0.085	7
	Serious games	0.118	4	0.119	4	0.128	4
	Storytelling	0.096	6	0.096	6	0.104	5
	Teamwork	0.175	1	0.174	1	0.146	2
	Training	0.066	8	0.066	8	0.074	8
	Volunteering activities	0.117	5	0.117	5	0.087	6

Table 5
Sensitivity to employees' psychological profile.

Sub-criteria	Professional Development Programs	Alpha value (α)		
		0.25	0.5	0.75
Interpersonal trust	Apprenticeship	0.044 (2.33 %)	0.043	0.043 (0.00 %)
	Communities of practice	0.188 (−4.08 %)	0.196	0.208 (6.12 %)
	Enterprise social networks	0.145 (2.11 %)	0.142	0.137 (−3.52 %)
	Mentoring	0.046 (2.22 %)	0.045	0.043 (−4.44 %)
	Serious games	0.106 (−14.52 %)	0.124	0.154 (24.19 %)
	Storytelling	0.117 (23.16 %)	0.095	0.060 (−36.84 %)
	Teamwork	0.174 (−8.90 %)	0.191	0.218 (14.14 %)
	Training	0.054 (10.20 %)	0.049	0.042 (−14.29 %)
	Volunteering activities	0.126 (10.53 %)	0.114	0.096 (−15.79 %)
	Volunteering activities	0.119 (4.39 %)	0.114	0.089 (−21.93 %)
Intrinsic motivation	Apprenticeship	0.038 (−11.63 %)	0.043	0.072 (67.44 %)
	Communities of practice	0.190 (−3.06 %)	0.196	0.226 (15.31 %)
	Enterprise social networks	0.147 (3.52 %)	0.142	0.117 (−17.61 %)
	Mentoring	0.041 (−8.89 %)	0.045	0.068 (51.11 %)
	Serious games	0.126 (1.61 %)	0.124	0.117 (−5.65 %)
	Storytelling	0.096 (1.05 %)	0.095	0.086 (−9.47 %)
	Teamwork	0.197 (3.14 %)	0.191	0.159 (−16.75 %)
	Training	0.046 (−6.12 %)	0.049	0.066 (34.69 %)
	Volunteering activities	0.119 (4.39 %)	0.114	0.089 (−21.93 %)
	Volunteering activities	0.114 (0.00 %)	0.114	0.119 (4.39 %)
Self-efficacy	Apprenticeship	0.052 (20.93 %)	0.043	0.030 (−30.23 %)
	Communities of practice	0.211 (7.65 %)	0.196	0.172 (−12.24 %)
	Enterprise social networks	0.128 (−9.86 %)	0.142	0.164 (15.49 %)
	Mentoring	0.049 (8.89 %)	0.045	0.039 (−13.33 %)
	Serious games	0.144 (16.13 %)	0.124	0.093 (−25.00 %)
	Storytelling	0.069 (−27.37 %)	0.095	0.136 (43.16 %)
	Teamwork	0.199 (4.19 %)	0.191	0.178 (−6.81 %)
	Training	0.051 (4.08 %)	0.049	0.047 (−4.08 %)
	Volunteering activities	0.097 (−14.91 %)	0.114	0.141 (23.68 %)
	Volunteering activities	0.114 (0.00 %)	0.114	0.119 (4.39 %)
Self-regulation skills	Apprenticeship	0.042 (−2.33 %)	0.043	0.061 (41.86 %)
	Communities of practice	0.196 (0.00 %)	0.196	0.194 (−1.02 %)
	Enterprise social networks	0.145 (2.11 %)	0.142	0.103 (−27.46 %)
	Mentoring	0.045 (0.00 %)	0.045	0.042 (−6.67 %)
	Serious games	0.122 (−1.61 %)	0.124	0.152 (22.58 %)
	Storytelling	0.096 (1.05 %)	0.095	0.081 (−14.74 %)
	Teamwork	0.192 (0.52 %)	0.191	0.179 (−6.28 %)
	Training	0.048 (−2.04 %)	0.049	0.070 (42.86 %)
	Volunteering activities	0.114 (0.00 %)	0.114	0.119 (4.39 %)
	Volunteering activities	0.114 (0.00 %)	0.114	0.119 (4.39 %)

- job satisfaction (the overall change varies from −2.05 %, when $\alpha = 0.25$ –5.11 %, when $\alpha = 0.75$).

On the other hand, they are inhibited by.

- collaboration (the overall change varies from 6.89 %, when $\alpha = 0.25$ to −35.34 %, when $\alpha = 0.75$),
- inclusiveness (the overall change varies from 3.89 %, when $\alpha = 0.25$ to −8.40 %, when $\alpha = 0.75$),
- supervisory skills (the overall change varies from 0.48 %, when $\alpha = 0.25$ to −15.20 %, when $\alpha = 0.75$).

Last but not least, the sensitivity analysis of structural capital (Table 7) proves that professional development programs are enhanced by company's size (the overall change varies from −1.71 %, when $\alpha = 0.25$ –51.91 %, when $\alpha = 0.75$), organizational culture (varies from −19.86 %, when $\alpha = 0.25$ –29.07 %, when $\alpha = 0.75$), and organizational structure (varies from −1.26 %, when $\alpha = 0.25$ –37.72 %, when $\alpha = 0.75$) and inhibited by organizational memory (varies from 14.13 %, when $\alpha = 0.25$ to −29.68 %, when $\alpha = 0.75$) and technology (varies from 7.77 %, when $\alpha = 0.25$ to −32.89 %, when $\alpha = 0.75$).

- apprenticeship is fostered by the enhancements in commitment, job design, company's size, organizational culture, and organizational structure;
- online communities of practice are positively influenced by the enhancements in collaboration, commitment, reward management, job satisfaction, organizational culture, and organizational memory;
- enterprise social networks are enhanced by collaboration, inclusiveness, organizational memory and technology;

- mentoring is fostered by commitment, job design, reward management, company's size, organizational culture, organizational memory, and organizational structure;
- serious games are positively influenced by the enhancements in collaboration, inclusiveness, job design, and technology;
- storytelling is enhanced by commitment, job design, reward management, job satisfaction, company's size, and organizational memory;
- teamwork is fostered by collaboration, commitment, job satisfaction, and organizational culture;
- training is positively influenced by the enhancements in job design, reward management, company's size, organizational culture, and organizational structure;
- volunteering activities are enhanced by collaboration, inclusiveness, and organizational culture.

5. Discussion

A four-stage methodology, which combines the qualitative approach of the Delphi study with the quantitative approach of ANP, was proposed and tested in the largest companies from the Romanian manufacture of coke and refined petroleum products industry.

These findings have both theoretical and practical implications. At the theoretical level, they provide a nexus between human resource management and operations management by showing how a methodology based on ANP that is frequently used in operations management area can be applied for solving a human resource issue. Furthermore, they extend the literature regarding Industry 5.0 by emphasizing both the professional development programs that enhance employees' reskilling and upskilling and also the factors that influence them.

Table 6
Sensitivity to human capital.

Sub-criteria	Professional Development Programs	Alpha value (α)		
		0.25	0.5	0.75
Collaboration	Apprenticeship	0.077 (5.48 %)	0.073	0.052 (−28.77 %)
	Communities of practice	0.117 (−2.50 %)	0.120	0.138 (15.00 %)
	Enterprise social networks	0.117 (−4.10 %)	0.122	0.147 (20.49 %)
	Mentoring	0.113 (6.60 %)	0.106	0.065 (−38.68 %)
	Serious games	0.106 (0.00 %)	0.106	0.107 (0.94 %)
	Storytelling	0.089 (3.49 %)	0.086	0.070 (−18.60 %)
	Teamwork	0.163 (−3.55 %)	0.169	0.201 (18.93 %)
	Training	0.097 (5.43 %)	0.092	0.068 (−26.09 %)
Commitment	Volunteering activities	0.121 (−3.97 %)	0.126	0.153 (21.43 %)
	Apprenticeship	0.070 (−4.11 %)	0.073	0.102 (39.73 %)
	Communities of practice	0.119 (−0.83 %)	0.120	0.128 (6.67 %)
	Enterprise social networks	0.126 (3.28 %)	0.122	0.092 (−24.59 %)
	Mentoring	0.097 (−8.49 %)	0.106	0.176 (66.04 %)
	Serious games	0.110 (3.77 %)	0.106	0.070 (−33.96 %)
	Storytelling	0.081 (−5.81 %)	0.086	0.122 (41.86 %)
	Teamwork	0.169 (0.00 %)	0.169	0.171 (1.18 %)
Executive skills	Training	0.096 (4.35 %)	0.092	0.065 (−29.35 %)
	Volunteering activities	0.132 (4.76 %)	0.126	0.074 (−41.27 %)
	Apprenticeship	0.074 (1.37 %)	0.073	0.055 (−24.66 %)
	Communities of practice	0.122 (1.67 %)	0.120	0.085 (−29.17 %)
	Enterprise social networks	0.123 (0.82 %)	0.122	0.085 (−30.33 %)
	Mentoring	0.104 (−1.89 %)	0.106	0.141 (33.02 %)
	Serious games	0.107 (0.94 %)	0.106	0.069 (−34.91 %)
	Storytelling	0.085 (−1.16 %)	0.086	0.109 (26.74 %)
Inclusiveness	Teamwork	0.168 (−0.59 %)	0.169	0.203 (20.12 %)
	Training	0.092 (0.00 %)	0.092	0.093 (1.09 %)
	Volunteering activities	0.124 (−1.59 %)	0.126	0.159 (26.19 %)
	Apprenticeship	0.075 (2.74 %)	0.073	0.069 (−5.48 %)
	Communities of practice	0.126 (5.00 %)	0.120	0.105 (−12.50 %)
	Enterprise social networks	0.113 (−7.38 %)	0.122	0.145 (18.85 %)
	Mentoring	0.110 (3.77 %)	0.106	0.095 (−10.38 %)
	Serious games	0.096 (−9.43 %)	0.106	0.131 (23.58 %)
Job design	Storytelling	0.096 (11.63 %)	0.086	0.061 (−29.07 %)
	Teamwork	0.178 (5.33 %)	0.169	0.146 (−13.61 %)
	Training	0.098 (6.52 %)	0.092	0.077 (−16.30 %)
	Volunteering activities	0.108 (−14.29 %)	0.126	0.172 (36.51 %)
	Apprenticeship	0.068 (−6.85 %)	0.073	0.147 (101.37 %)
	Communities of practice	0.123 (2.50 %)	0.120	0.081 (−32.50 %)
	Enterprise social networks	0.124 (1.64 %)	0.122	0.085 (−30.33 %)
	Mentoring	0.103 (−2.83 %)	0.106	0.138 (30.19 %)
Reward management	Serious games	0.105 (−0.94 %)	0.106	0.121 (14.15 %)
	Storytelling	0.084 (−2.33 %)	0.086	0.116 (34.88 %)
	Teamwork	0.172 (1.78 %)	0.169	0.129 (−23.67 %)
	Training	0.092 (0.00 %)	0.092	0.104 (13.04 %)
	Volunteering activities	0.129 (2.38 %)	0.126	0.077 (−38.89 %)
	Apprenticeship	0.073 (0.00 %)	0.073	0.072 (−1.37 %)
	Communities of practice	0.120 (0.00 %)	0.120	0.136 (13.33 %)
	Enterprise social networks	0.123 (0.82 %)	0.122	0.105 (−13.93 %)
Supervisory skills	Mentoring	0.104 (−1.89 %)	0.106	0.133 (25.47 %)
	Serious games	0.106 (0.00 %)	0.106	0.097 (−8.49 %)
	Storytelling	0.085 (−1.16 %)	0.086	0.101 (17.44 %)
	Teamwork	0.171 (1.18 %)	0.169	0.143 (−15.38 %)
	Training	0.092 (0.00 %)	0.092	0.104 (13.04 %)
	Volunteering activities	0.127 (0.79 %)	0.126	0.109 (−13.49 %)
	Apprenticeship	0.073 (0.00 %)	0.073	0.069 (−5.48 %)
	Communities of practice	0.120 (0.00 %)	0.120	0.125 (4.17 %)
Job satisfaction	Enterprise social networks	0.125 (2.46 %)	0.122	0.074 (−39.34 %)
	Mentoring	0.106 (0.00 %)	0.106	0.096 (−9.43 %)
	Serious games	0.106 (0.00 %)	0.106	0.105 (−0.94 %)
	Storytelling	0.086 (0.00 %)	0.086	0.083 (−3.49 %)
	Teamwork	0.167 (−1.18 %)	0.169	0.214 (26.63 %)
	Training	0.092 (0.00 %)	0.092	0.092 (0.00 %)
	Volunteering activities	0.125 (−0.79 %)	0.126	0.142 (12.70 %)
	Apprenticeship	0.076 (4.11 %)	0.073	0.064 (−12.33 %)
	Communities of practice	0.116 (−3.33 %)	0.120	0.134 (11.67 %)
	Enterprise social networks	0.122 (0.00 %)	0.122	0.122 (0.00 %)
	Mentoring	0.109 (2.83 %)	0.106	0.096 (−9.43 %)
	Serious games	0.109 (2.83 %)	0.106	0.097 (−8.49 %)
	Storytelling	0.083 (−3.49 %)	0.086	0.094 (9.30 %)
	Teamwork	0.167 (−1.18 %)	0.169	0.174 (2.96 %)
	Training	0.079 (−14.13 %)	0.092	0.131 (42.39 %)
	Volunteering activities	0.139 (10.32 %)	0.126	0.087 (−30.95 %)

Table 7
Sensitivity to structural capital.

Sub-criteria	Professional Development Programs	Alpha value (α)		
		0.25	0.5	0.75
Company's size	Apprenticeship	0.077 (−1.28 %)	0.078	0.103 (32.05 %)
	Communities of practice	0.181 (0.56 %)	0.180	0.152 (−15.56 %)
	Enterprise social networks	0.140 (0.72 %)	0.139	0.091 (−34.53 %)
	Mentoring	0.096 (−1.03 %)	0.097	0.121 (24.74 %)
	Serious games	0.139 (1.46 %)	0.137	0.090 (−34.31 %)
	Storytelling	0.111 (−0.89 %)	0.112	0.128 (14.29 %)
	Teamwork	0.118 (0.00 %)	0.118	0.137 (16.10 %)
	Training	0.080 (−1.23 %)	0.081	0.118 (45.68 %)
	Volunteering activities	0.058 (0.00 %)	0.058	0.060 (3.45 %)
	Apprenticeship	0.073 (−6.41 %)	0.078	0.084 (7.69 %)
Organizational culture	Communities of practice	0.175 (−2.78 %)	0.180	0.187 (3.89 %)
	Enterprise social networks	0.158 (13.67 %)	0.139	0.111 (−20.14 %)
	Mentoring	0.095 (−2.06 %)	0.097	0.099 (2.06 %)
	Serious games	0.161 (17.52 %)	0.137	0.105 (−23.36 %)
	Storytelling	0.114 (1.79 %)	0.112	0.110 (−1.79 %)
	Teamwork	0.105 (−11.02 %)	0.118	0.137 (16.10 %)
	Training	0.073 (−9.88 %)	0.081	0.092 (13.58 %)
	Volunteering activities	0.046 (−20.69 %)	0.058	0.076 (31.03 %)
	Apprenticeship	0.081 (3.85 %)	0.078	0.071 (−8.97 %)
	Communities of practice	0.170 (−5.56 %)	0.180	0.204 (13.33 %)
Organizational memory	Enterprise social networks	0.134 (−3.60 %)	0.139	0.149 (7.19 %)
	Mentoring	0.095 (−2.06 %)	0.097	0.101 (4.12 %)
	Serious games	0.141 (2.92 %)	0.137	0.130 (−5.11 %)
	Storytelling	0.104 (−7.14 %)	0.112	0.132 (17.86 %)
	Teamwork	0.121 (2.54 %)	0.118	0.112 (−5.08 %)
	Training	0.090 (11.11 %)	0.081	0.059 (−27.16 %)
	Volunteering activities	0.065 (12.07 %)	0.058	0.043 (−25.86 %)
	Apprenticeship	0.075 (−3.85 %)	0.078	0.128 (64.10 %)
	Communities of practice	0.183 (1.67 %)	0.180	0.137 (−23.89 %)
	Enterprise social networks	0.139 (0.00 %)	0.139	0.128 (−7.91 %)
Organizational structure	Mentoring	0.096 (−1.03 %)	0.097	0.113 (16.49 %)
	Serious games	0.139 (1.46 %)	0.137	0.120 (−12.41 %)
	Storytelling	0.112 (0.00 %)	0.112	0.114 (1.79 %)
	Teamwork	0.118 (0.00 %)	0.118	0.118 (0.00 %)
	Training	0.080 (−1.23 %)	0.081	0.096 (18.52 %)
	Volunteering activities	0.059 (1.72 %)	0.058	0.047 (−18.97 %)
	Apprenticeship	0.083 (6.41 %)	0.078	0.055 (−29.49 %)
	Communities of practice	0.188 (4.44 %)	0.180	0.145 (−19.44 %)
	Enterprise social networks	0.126 (−9.35 %)	0.139	0.194 (39.57 %)
	Mentoring	0.101 (4.12 %)	0.097	0.076 (−21.65 %)
Technology	Serious games	0.115 (−16.06 %)	0.137	0.233 (70.07 %)
	Storytelling	0.119 (6.25 %)	0.112	0.083 (−25.89 %)
	Teamwork	0.126 (6.78 %)	0.118	0.084 (−28.81 %)
	Training	0.081 (0.00 %)	0.081	0.081 (0.00 %)
	Volunteering activities	0.061 (5.17 %)	0.058	0.048 (−17.24 %)

Thus, with regard to the organizational factors, it can be stated that.

The results are in line with [33] by showing that teamwork is the most appropriate professional development program for employees' reskilling and upskilling. On the same time, they contradict the findings of [21] which describe apprenticeship as the major mean for employees' reskilling and upskilling, in the Industry 5.0 context. This may happen due to the fact that apprenticeship usually focuses on developing technical skills and not the soft skills (critical thinking, creativity, communication, collaboration) that are required by Industry 5.0.

Furthermore, the current findings complement the work of [46,47] by quantifying the influence of employees' psychological profile on the professional development programs. Thus, they emphasize that interpersonal trust and self-efficacy inhibit skills development through professional development programs while intrinsic motivation and self-regulation skills act as stimulus. Hence, if the importance of interpersonal trust increases by 50 %, the overall change in professional development programs sums −30.43 %. On the other hand, if the importance of intrinsic motivation increases by 50 %, the overall change sums 97.14 %. While intrinsic motivation acts as a stimulus for all the

professional development programs that foster skills development, interpersonal trust only enhances those programs that focus on developing communication and collaboration, namely online communities of practice and serious games.

In addition, this research complements the works of [16,71] by quantifying the influence of job design on the professional development programs. Thus, if its importance decreases by 50 %, the overall change in professional development programs equals −4.65 %; if the importance increases by 50 %, the overall change sums up to 68.25 %. The change occurs due to the fact that job design acts as a stimulus for apprenticeship, mentoring, storytelling, serious games and training, and as an inhibitor for online communities of practice, enterprise social networks, teamwork and volunteering activities. In other words, it facilitates the professional development programs that concentrate on the development of hard skills, and it inhibits those that foster the development and use of soft skills. Nevertheless, further research is required in order to determine which skills are developed.

Additionally, the research findings are in line with [22,23] who state

Table 8

Factors that enhance employees' reskilling and upskilling through professional development programs.

Professional development programs Factors	Apprenticeship	Mentoring	Training	Teamwork	Storytelling	Communities of practice	Enterprise social network	Serious games	Volunteering activities
Interpersonal trust	–	–	–	+	–	+	–	+	–
Self-efficacy	–	–	–	–	+	–	+	–	+
Intrinsic motivation	+	+	+	–	–	+	–	–	–
Inclusiveness	–	–	–	–	–	–	+	+	+
Job satisfaction	–	–	+	+	+	+	+	–	–
Collaboration	–	–	–	+	–	+	+	+	+
Organizational culture	+	+	+	+	–	+	–	–	+
Organizational memory	–	+	–	–	+	+	+	–	–
Technology	–	–	+	–	–	–	+	+	–

that organizational culture and commitment have a strong influence on mentoring. Hence, if the importance of organizational culture and commitment increases by 50 %, the change in mentoring equals 2.06 %, respectively 66.04 %. Furthermore, both factors act as stimulus for the professional development programs especially, for apprenticeship, mentoring, teamwork and online communities of practice.

Nevertheless, the current research not only has theoretical implications but also practical ones. Thus, it helps managers to decide which professional development programs to implement for employees' reskilling and upskilling by providing them a viable tool for programs' prioritizing. As the results have shown, employees' reskilling and upskilling can be enhanced through teamwork, online communities of practice, and enterprise social networks. However, this is not a successful recipe and must be adapted to company's characteristics. This study offers a methodology that can be easily customized, based on company's needs and interests. Different results can be obtained if this methodology is applied in different industries and/or companies since it reflects the position of the policy-makers. As aforementioned, if the managers focus more on employees' commitment than all the other factors, skills development should be enhanced through mentoring. If they focus on inclusiveness, the wright choice will be to develop volunteering activities. If managers decide to concentrate on organizational memory, they could enhance employees' reskilling and upskilling by developing online communities of practice and stimulating the use of enterprise social networks and storytelling.

Despite these insights, the current research is limited by the fact that it reflects the perspective of human resource managers and not the end-users of the professional development programs. Different results can be obtained if employees' opinion is taken into consideration. Furthermore, this study focuses on the professional development programs as a whole and not on the skill set that they develop. Further research should be developed in this area in order to facilitate decision-making. Last but not least, this study brings forward the most appropriate professional development programs for the Romanian manufacture of coke and refined petroleum products industry. Different results can be obtained for different industries and countries.

6. Conclusion

The current research aimed at developing a multi-criteria decision-making methodology, based on ANP, for selecting the appropriate professional development programs for enhancing employees' reskilling and upskilling, in the Industry 5.0 context.

The results have shown that teamwork, online communities of practice and enterprise social networks are among the most appropriate professional development programs for fostering employees' reskilling and upskilling. Furthermore, the most influential factors belong to employees' psychological profile (namely, interpersonal trust, self-efficacy, and intrinsic motivation) and human capital (especially, inclusiveness, job satisfaction and collaboration), allowing the customization of the professional development programs and making sure that the content is

centered on employees' needs and adapted to their characteristics. Nevertheless, the influence of structural capital (namely, organizational culture, organizational memory and technology) should not be neglected since it provides the infrastructure for decision-making and programs' implementation. Table 8 synthesizes the influence of each of these factors.

Starting from these and the research limitations, further research could be developed regarding: (i) the prioritization of the professional development programs for specific groups of skills development; and (ii) employees' preferences in terms of professional development programs.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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